

EXTENDS *Naturals*

VARIABLES *Numbers*

$IsEven(n) \triangleq (n \% 2 = 0)$
 $IsDivisibleBy4(n) \triangleq (n \% 4 = 0)$
 $IsDivisibleBy2(n) \triangleq (n \% 2 = 0)$

Fails, because not correct

$Example1 \triangleq \forall n \in Numbers : IsDivisibleBy4(n) \equiv IsEven(n)$

$Example2 \triangleq \forall n \in Numbers : IsEven(n) \equiv IsDivisibleBy2(n)$

$Init \triangleq Numbers = 0 \dots 10$
 $Next \triangleq UNCHANGED\ Numbers$
 $Spec \triangleq Init \wedge \Box[Next]_{Numbers}$